
CHAPTER 1

TRICKERY

1 Some Telescoping Trickery

Example : $s = 1(1!) + 2(2!) + \cdots + n(n!) = (n+1)! - 1$

Simple application of **telescoping** trick. Write $j(j!) = (j+1-1)j! = (j+1)j! - j! = (j+1)! - j!$ which gives us $s = \sum_{j=1}^n (j)(j!) = \sum_{j=1}^n (j+1)! - j!$ which is just $(2! - 1!) + (3! - 2!) + \cdots + (n! - (n-1)!) + (n+1! - n!) = (n+1)! - 1$ ■